

Introduction

CS341: Computer Graphics

January 2022



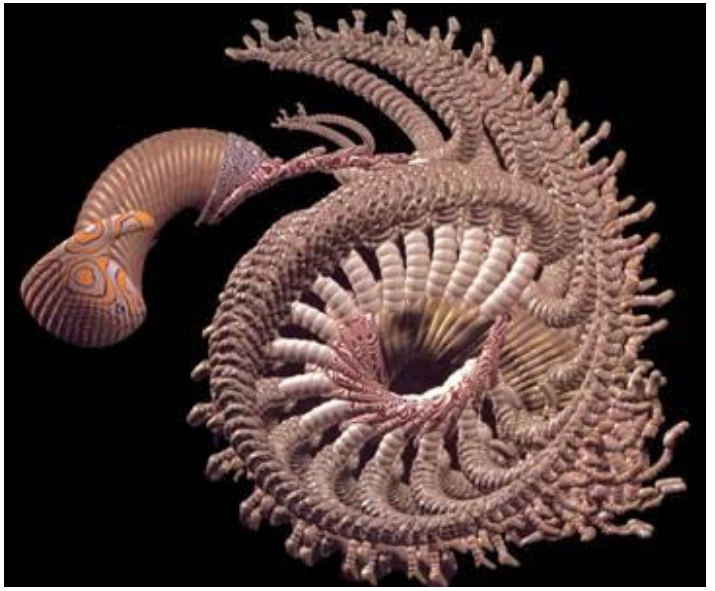
Qassim University
جامعة القصيم
College Of Computer | كلية الحاسب

The Course

- Trivia
  - Text Book
  - *Computer Graphics with OpenGL, 4th Edition, Hearn, D. D.; Baker, M. P.; and Carithers, W. 2014*
  - Supplemental Texts
  - Fundamentals of Computer Graphics, Peter Shirley, A.K. Peters, 2002
  - Interactive Computer Graphics: A top down approach using OpenGL, Edward Angel, Addison Wesley, 3rd Edition, 2003
  - Lecture Notes and Handouts
- Class page
  - Keep checking your blackboard (lms.qu.edu.sa)

What is Computer Graphics?

Computer graphics is commonly understood to mean the **creation**, **storage,** and **manipulation** of models and images.



William Latham:
SIGGRAPH 1992

Sources of Models

- Models come from fields like:
  - physical,
  - mathematical,
  - artistic,
  - biological, and
  - even conceptual (abstract) structures.



More Easily

- Computer graphics
  - generating **2D images** of a **3D world** represented in a computer
- Main tasks
  - ***modeling***: creating and representing the geometry of objects in the 3D world
  - ***rendering***: generating 2D images of the objects
  - ***animation***: describing how objects change in time
- The term "computer graphics" was coined in 1960 by William Fetter.

Computer Graphics Goals

- Synthetic images indistinguishable from reality
- Practical, scientifically sound, in real time



Interactive Computer Graphics

- User controls contents, structure, and appearance of objects and their displayed images via rapid visual **feedback**
- Basic components of an interactive graphics system
  - **input** (e.g., mouse, tablet and stylus, force feedback device, scanner, live video streams…)
  - **processing** (and storage)
  - **display/output** (e.g., screen, paper-based printer, video recorder…)



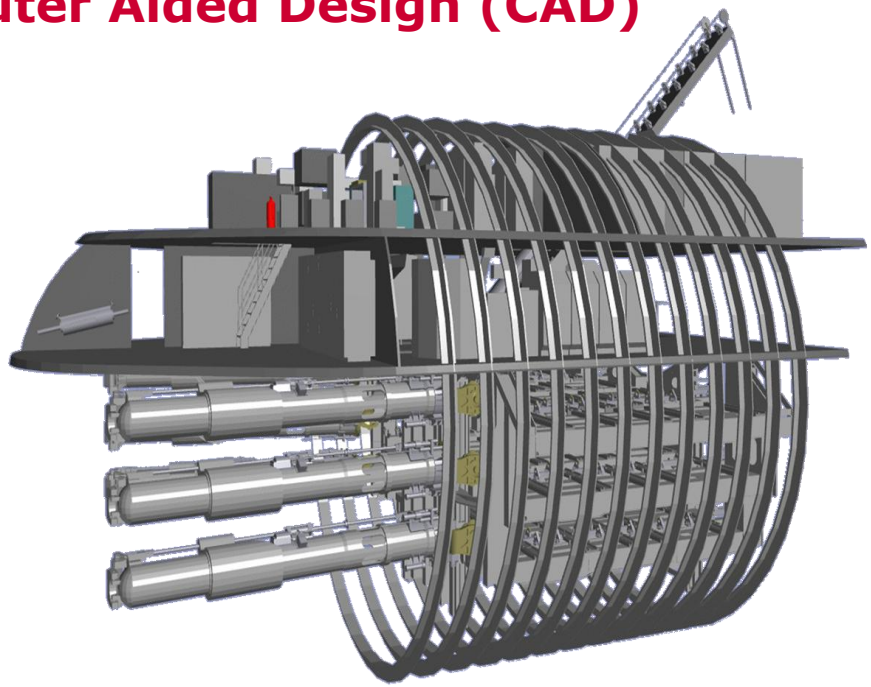
First truly interactive graphics system, Sketchpad (MIT 1963)

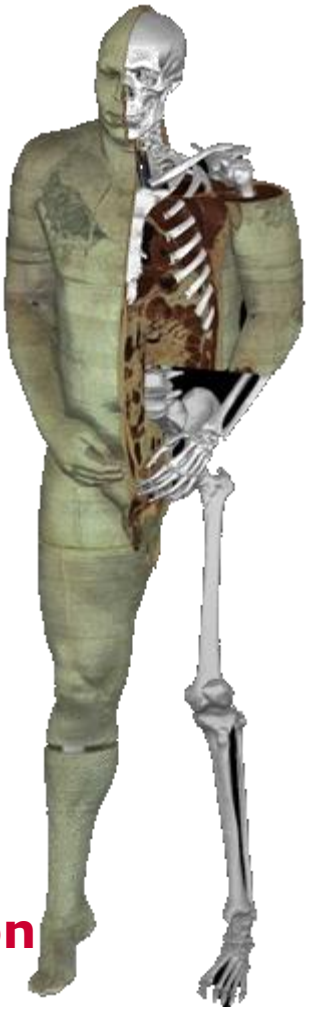
Applications of Computer Graphics



Graphics Applications

Computer Aided Design (CAD)





Medical Visualization

Graphics Applications

Entertainment: Games

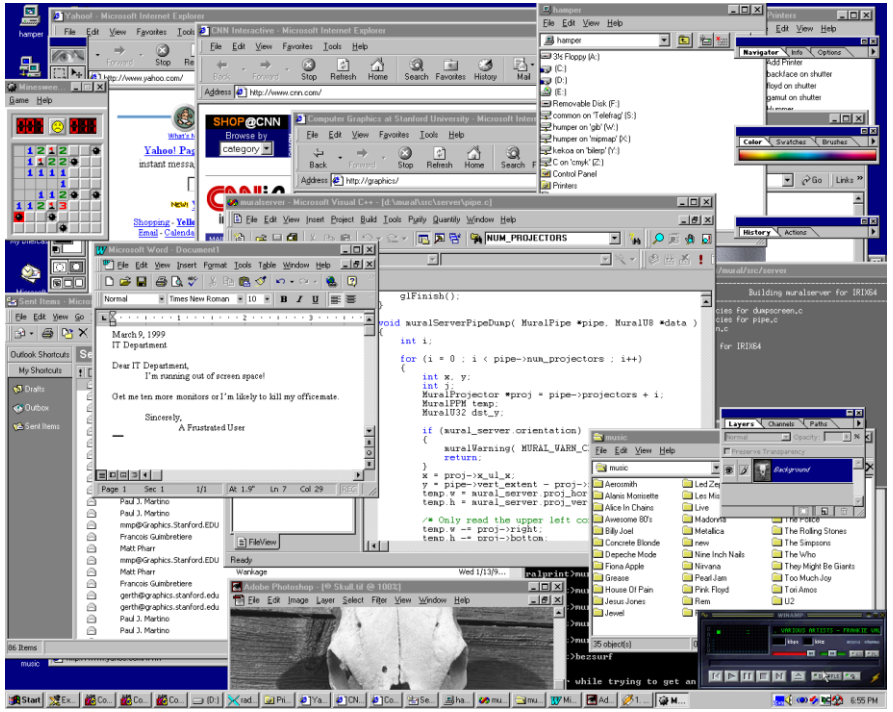


Entertainment: Cinema



Graphics Applications

Everyday Use: user interface

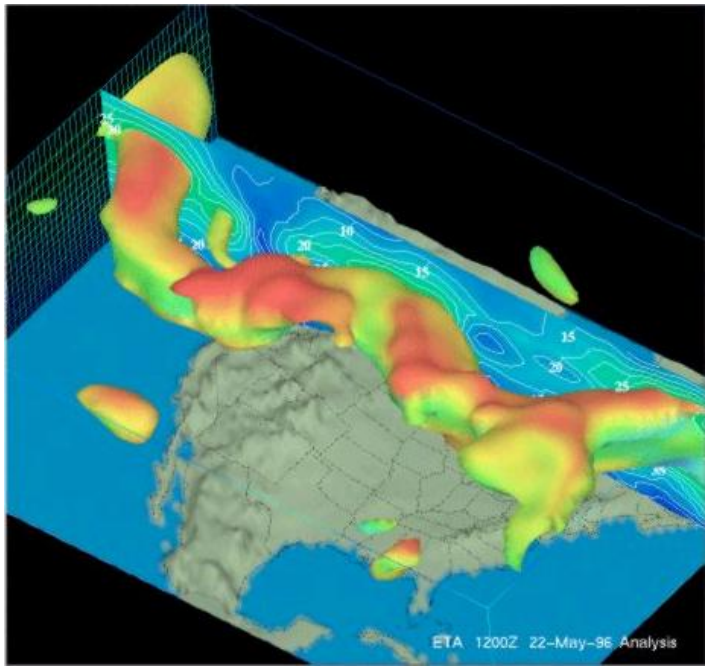


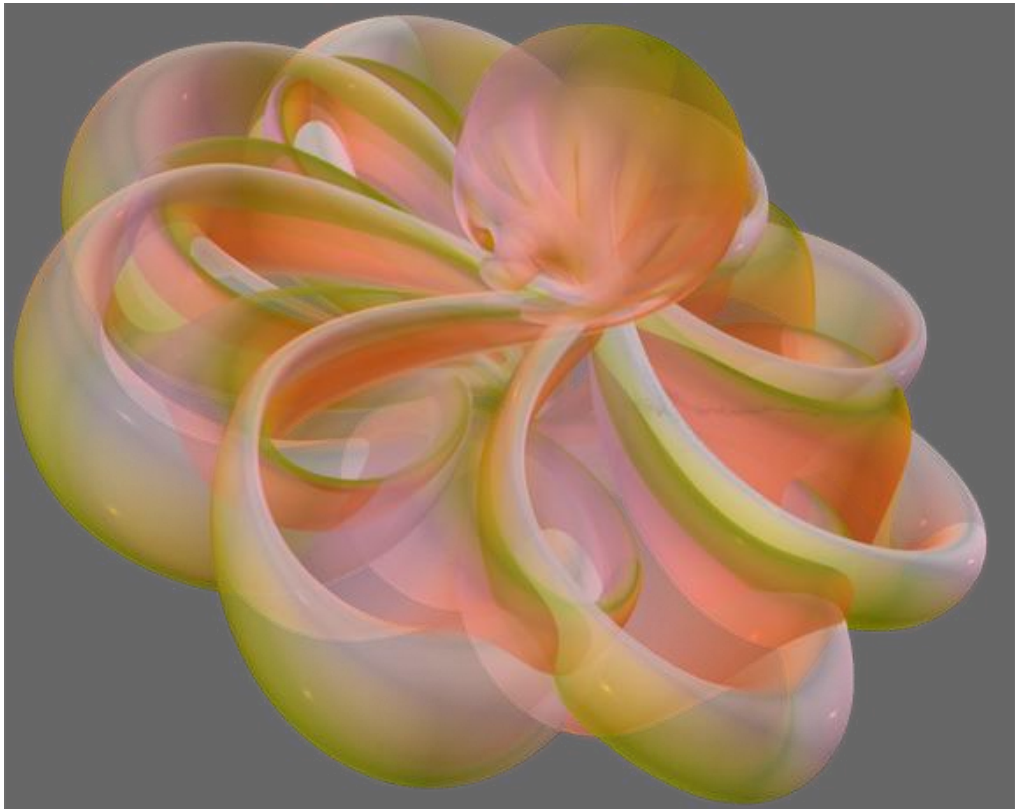
Everyday Use: large-screen interaction



Graphics Applications

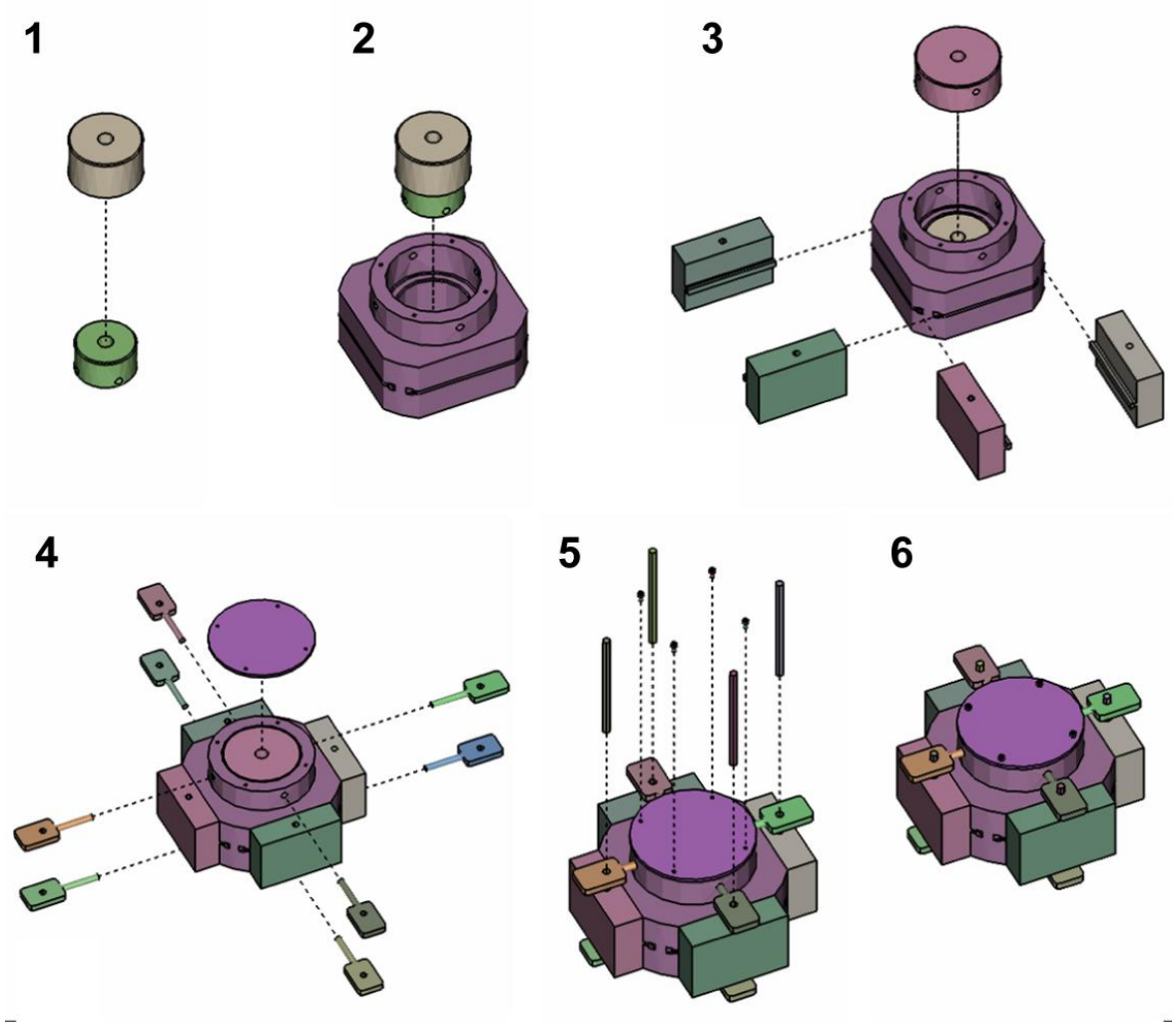
Simulation





Education

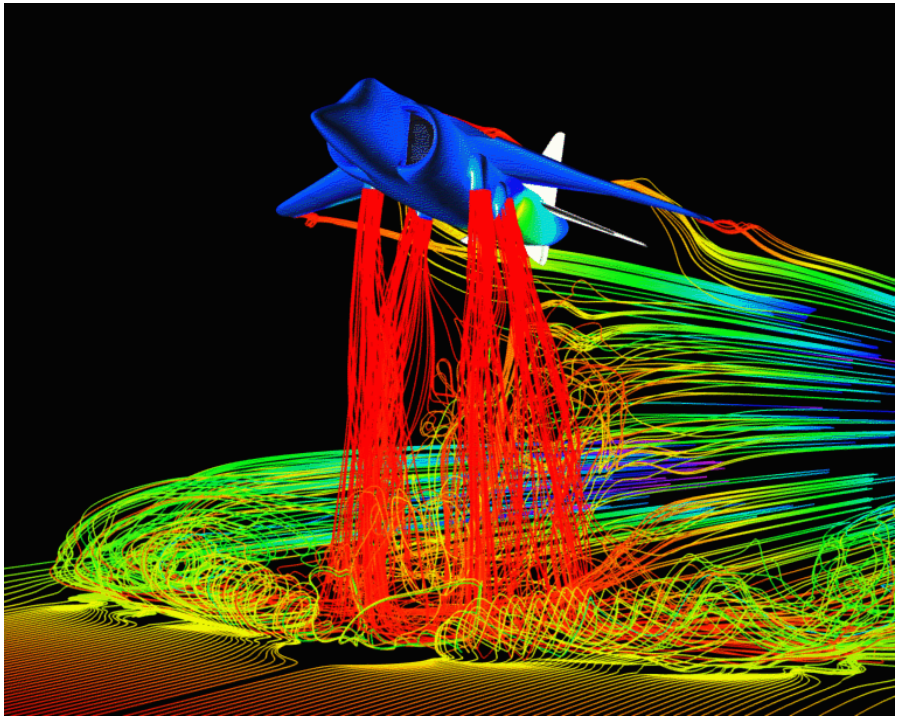
Graphics Applications

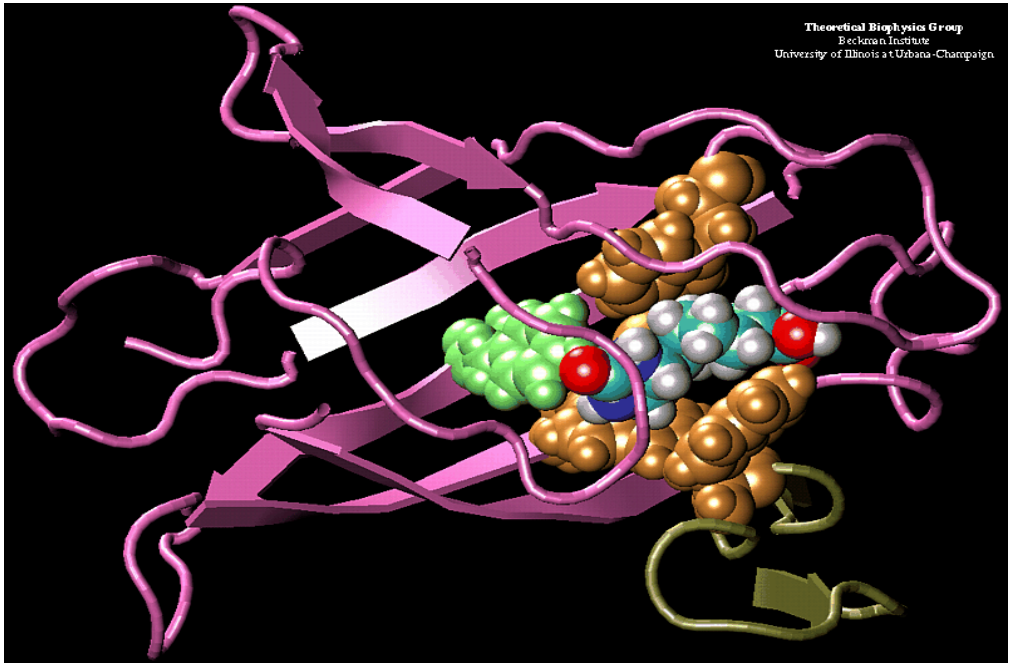


Training

Graphics Applications

Scientific Visualization





So…

- Why study Computer Graphics?
  - It's **COOL**…
  - It's **HOT!**
  - It's **FUN**



This course is about …

We will cover

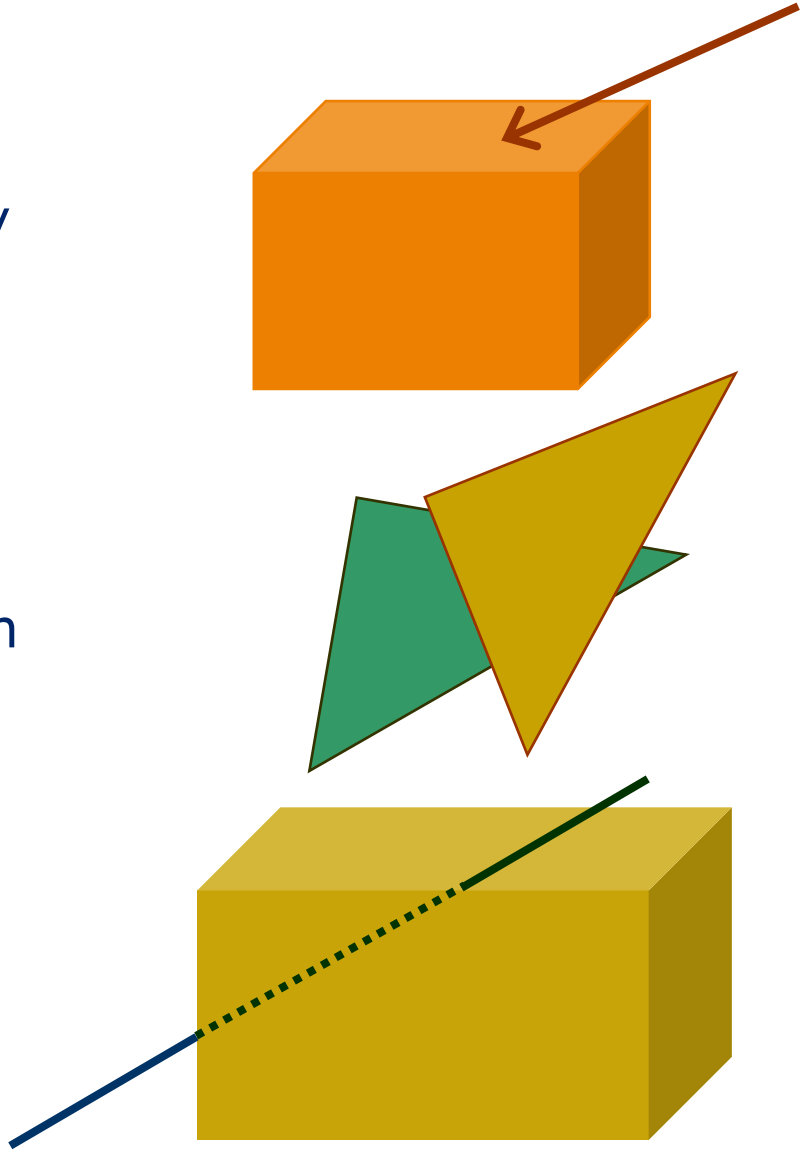
- Graphics programming algorithms
- Graphics data structures
- Color and human vision
- Applied geometry and modeling
- Graphics API

It is NOT

- Paint and Imaging packages (Adobe Photoshop)
- CAD packages (AutoCAD)
- Rendering packages (Lightscape)
- Modeling packages (3D Studio MAX)
- Animation packages (Digimation)
- Graphics Modeling and Languages (RenderMan)
- Game Programming (GPU)
- User Interface Design
- Image Processing

So we face questions…

- Given a cube and a ray, does the ray strike the cube? Where? What if the ray reflects off?
- Which part of the triangle is hidden?
- Given a polyhedron and a line, which part of the line lie within the polyhedron?



The Topics

- **Graphics Hardware**
  - Display devices
  - Input devices
- **Modeling**
  - Hierarchical Models
  - Animations
  - Fractals
- **Objects & Transformations**
  - Linear algebra, Affine geometry and Coordinate systems
  - Basic transformations: rotation, translation, scaling
  - Composite transformations
- **Viewing and Projection**
  - Orthographic projection
  - Perspective projection
- **Lighting and shading**
  - Light sources
  - Ambient, diffuse, and specular reflection
  - Normal vectors
  - Phong shading
  - Gouraud shading
  - Radiosity

The Topics

- **Curves and Surfaces**
  - Hermite, Bezier curves and surfaces
  - Splines
- **Rendering Algorithms**
  - 2D and 3D Clipping
  - Line drawing
  - Scan conversion
  - Antialiasing
  - Hidden surface removal
    - z-buffer method, BSP trees, Octrees
- **Texture mapping**
  - Bump mapping
  - Environment (reflection) mapping
- **Color models**
- **Computer animation**

PLUS...